Whose Meaning Is the Default? Loss, Violation, and Assumption in LLM Keyword Expansion for AAC

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Abstract

Large language models can expand a few keywords into a fluent sentence, which suits augmentative and alternative communication (AAC). Yet a fluent expansion can change what the person meant: it can drop a refusal, change who acts, harden a hedge, or soften a deliberately plain request. We separate three sources of such changes by construction. *Loss* is information the keywords never carried; *Violation* is a cue the model received and changed; *Assumption* is a commitment the model made where the input left it open. Judges extract labels from outputs alone, blind to model, prompt and input channel, and attribution is computed deterministically. In **[s1_n_messages]** crowd-written AAC messages, content-word keywords drop at least one explicit commitment cue in **[s1_loss_excl_agent]**% of messages, not counting pronouns. On 50 minimal-pair families and four models, bare keywords are almost always expanded to the affirmative, so the negated meaning costs an extra action that the affirmative never does. One commitment chip largely restores the cue, but models still add "please" to plain directives typed in full, and further chips can undo a choice already made. We argue that defaults in AAC expansion are a policy choice and outline designs that make non-default meanings cheap.

Keywords

augmentative and alternative communication, large language models, text expansion, meaning preservation, user agency, evaluation

1. Introduction

Many people who use augmentative and alternative communication (AAC) compose by selecting a few words, because every selection costs time and effort [1, 2]. This includes many autistic people, some of whom use AAC all the time and some only when speech is unreliable or unavailable [3, 4]. Large language models (LLMs) make keyword expansion attractive: from `want go party` a model can produce a complete, fluent sentence in under a second [5, 2].

Fluency is not the same as saying what the person meant. The words a person leaves out are often the ones that carry their commitments: "not", "I" or "you", "maybe", a question mark, or the absence of "please". An expansion that turns "I don't want to go" into "I want to go" is fluent, and it says the opposite. AAC users and the people who work with them have named this risk directly. They want outputs they can edit and recognize as their own [1, 6], and they raise authorship as a governance issue [7, 8]. Research on AI-mediated communication shows that suggestions can shift what people write and even what they think [9, 10, 11].

We do not assume an "autistic style" of language, and we do not compare synthetic autistic and non-autistic text. Adult autistic perspectives point to individual and context-dependent choices, not to a single register [3, 12]. Our position is simpler and, we think, easier to defend: when a person explicitly refuses, negates, speaks directly, or hedges, that choice should stay theirs. A system can be helpful about grammar and still leave those choices alone.

Closest work and the gap. Several systems already give the person explicit control over intent. COMPA offers intent buttons and three starter phrases in conversation [13]. Mao et al. let people with aphasia choose request, question or statement before keyword expansion into three sentences [14].

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SocializeChat offers editable stance tags such as Agree and Hesitant [15], and SPICA combines personalization, multiple options and clarification [16]. Evaluation work is close as well. Choudhury et al. measure whether affect cues reach generated AAC sentences [17]; Lacosse et al. retain function words such as "not" and "you" and vary pragmatic context [18]; and Gorenshtein et al. benchmark semantic drift, including negation and recipient probes, under simulated decoder errors [19]. None of these studies, as far as we could inspect them, separates three things: information the input never carried, a cue that the model received and changed, and a commitment made when the input was ambiguous. None of them prices what it costs the person to get a non-default meaning. Without that separation, a single "fidelity" score cannot say whether to fix the input interface, the model, or the default.

Contributions. We report an offline, controlled audit of LLM keyword expansion for AAC.

1. **Attribution by construction.** Controlled input channels determine whether an error counts as Loss, Violation, or Assumption. Judges only extract what an output says; attribution is a deterministic comparison with the item specification (§3).
2. **The default asymmetry and its action cost.** Under ambiguous keywords, models commit to one reading of a minimal pair far more often than the other. We report how many extra input actions each reading costs, separately for each direction (§5.2).
3. **Cue interaction and encoding.** An explicit choice that a single chip preserves can be lost when further chips are added. We measure this effect against an inert padding control and against alternative encodings of the same commitments (§5.4).
4. **A validated label-extraction oracle.** We release the codebook, probe families and judge, validated against held-out human labels; autistic advisers reviewed the codebook (§5.6). [PENDING: write as a contribution only if the validation gate passes; otherwise move it to Method as a limitation]

Conversational agency depends on what, when and to whom someone can communicate [20]; we measure only whether explicit textual commitments survive expansion. We make no claims about user benefit, agency, or authenticity, which require the people who use AAC to judge them [21].

2. Related Work

Meaning and stance in generation. Negated keyword inputs still produce errors in constrained generation [22], and LLM summaries often harden hedged source claims [23]. Models rarely express uncertainty [24], tend to answer rather than ask when an input is ambiguous [25], and can drift toward agreement and majority writing styles [26, 27]. In everyday human requests, "please" is rare and does specific interactional work [28], so adding it is a choice rather than a neutral repair. Interpretation sets for ambiguous requests already exist outside AAC [29, 30]. Our suggestion-set comparison is therefore an evaluation in the AAC setting, not a new method.

Behavioral testing and LLM judges. Minimal pairs and directional expectations come from behavioral testing [31] and compositional constraint benchmarks [32]. LLM judges agree with people on some tasks but show position, verbosity and self-preference biases [33, 34], so we validate ours against held-out human labels before relying on it.

3. Attribution by Construction

A keyword expander maps an input x to an output sentence y. For a source message s, we care about its *commitments*: the values that s makes explicit on six dimensions. These are polarity, the agent of the main action (speaker, addressee, joint or third party), certainty, time and quantity, speech act (statement, question, request, refusal), and politeness or mitigation. A commitment is a property of the text, not of the author's private preferences.

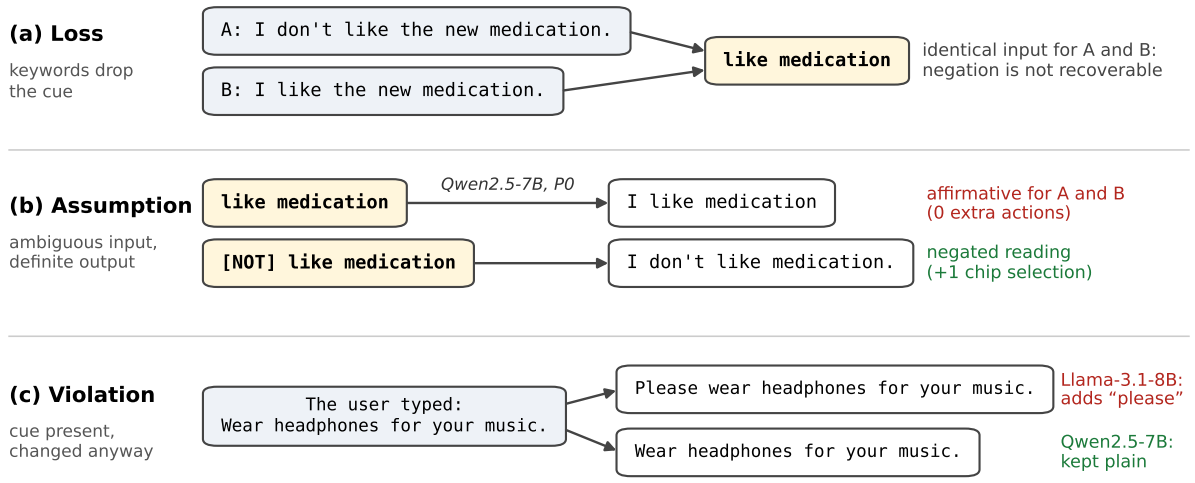


Figure 1: Three sources of meaning change, with real outputs from the main run. (a) Two source messages that differ only in negation map to the same keywords. (b) Given those keywords, the model commits to the affirmative reading for both; one `[NOT]` chip restores the negated reading. (c) Given the full typed sentence, one model adds "please" to a plain directive. Selection rule: the first test family of the dimension, prompt P0, Qwen2.5-7B. Panel (c) also shows the first model, in a fixed order, whose output adds "please"; Qwen2.5-7B kept the sentence plain. Script and selected rows: `figs/make_fig1.py`, `figs/fig1_examples.json`.

We build items as minimal pairs. A *family* has two source variants that differ on exactly one critical dimension and share the same content words, for example "I don't like the new medication" and "I like the new medication" (Fig. 1a). Both variants map to the same *bare* keywords (`like medication`), so the bare input cannot tell them apart. We then vary only what the model receives, the *channel*, and read off the error source from the channel:

- **Loss**: the channel does not carry the critical value. For probes this holds by design under the bare channel. For a corpus, we estimate it by checking whether each cue's carrier words survive compression (§4.1).
- **Violation**: the channel carries the critical cue, through a chip or the full source text, and the output's extracted value differs from the variant's value.
- **Assumption**: the channel is ambiguous between the two variants, and the output commits to a definite value anyway. Because both variants receive the identical input, the same output is "right" for one and "wrong" for the other. We therefore report the commitment rate and its direction, not an accuracy.

These are explanatory labels with separate denominators, not mutually exclusive causes. One case can involve both Loss (compression removed "not") and Assumption (the model then chose "like"). We never sum them into a single breakdown of causes. An improvement from the bare channel to the full source shows that the output depends on the input channel; it does not reveal the model's internal reasoning.

4. Method

4.1. S1: How often do keywords lose commitments?

We use AACText, **`[s1_n_messages]`** fictional messages written by crowd workers who imagined being AAC users (CC BY 4.0) [35]. These are not naturally occurring AAC messages, and we make no claim about any population of AAC users. A deterministic compressor keeps content words (nouns, proper

Table 1
The first test family of each scored dimension. The bare keywords are shared by both variants.

Dimension	Variant A	Variant B	Bare input	Chips A / B
Polarity	I don't like the new medication.	I like the new medication.	`like medication`	`[NOT]` / `[YES`
Agent	You left the stove on again.	I left the stove on again.	`leave stove on`	`[YOU]` / `[I]`
Certainty	I have a fever again.	I think I have a fever again.	`have fever`	`[SURE]` / `[MA`
Speech act	Can I bring my dog tomorrow?	I'm bringing my dog tomorrow.	`bring dog`	`[?]` / `[TELL`
Politeness	Wear headphones for your music, please.	Wear headphones for your music.	`wear headphones`	`[PLEASE]` / [

Table 2
Channels for variant A of the polarity family, with the action cost under our model. A chip costs one selection. A keyword costs one selection under word prediction, or its characters without it. Typed characters are shown for the full source.

Channel	Model input	Selections (chars)	Role in attribution
bare	`like medication`	2 (15)	ambiguous: Assumption
noncrit	`like new medication`	3 (19)	length control, still ambiguous
chip1	`[NOT] like medication`	3 (15 + 1 chip)	critical cue given: Violation
chipall	`[NOT] [TELL] [I] like medication`	5 (15 + 3 chips)	all commitments given: Violation
full	`The user typed: I don't like…`	– (32)	oracle: errors are Violation only

nouns, main verbs, adjectives, numerals), drops auxiliaries, keeps "please", and truncates to four keywords. A rule-based detector (spaCy plus lexicons) finds explicit cues for each dimension and checks whether their carrier tokens survive. Loss is the share of messages with at least one cue whose carrier is absent. We also compute the minimum number of chips that restores the detected values. Pronoun loss dominates the agent dimension and context can often supply the agent, so our headline excludes agent (Table 3). We checked the detector against a stratified sample of 400 messages. [PENDING: independent human check of detector labels]

4.2. S2: Probe families and channels

The held-out test set has 60 families, 10 per dimension, authored to a fixed specification. Each family records the two sources and their critical values, the shared bare keywords, one noncritical content word (for example `new`), the chips that disambiguate each variant, and the full set of commitments of each variant. Every variant has at least one chip; the "unmarked" variant gets an explicit counter-chip (`[YES]`, `[TELL]`, `[PLAIN]`), so neither variant is resolved for free. An independent reviewer checked every family against the specification before generation; nine were revised and none dropped. We developed the prompts and codebook on 24 pilot families and 12 development families, then froze them before running the test families. We do not score the ten time and quantity families in this paper: quantity loss is rare in S1, and our time labels are too coarse to score reliably. Table 1 shows one family per scored dimension.

Each variant is presented through five channels (Table 2). The *noncrit* channel adds the noncritical word. It controls for the length and position of the cue: if a chip helps only because the input got longer, a noncritical word should help equally. Chips are placed before the keywords in a fixed order, and keywords keep their source order.

Chips and cost. The chip vocabulary covers polarity (`[NOT]`, `[YES]`), speech act (`[?]`, `[ASK]`, `[REFUSE]`, `[TELL]`), certainty (`[MAYBE]`, `[SURE]`, `[MUST]`, `[WANT]`), agent (`[I]`, `[YOU]`, `[WE]`, `[THEY]`), time (`[PAST]`, `[NOW]`, `[LATER]`), exact numbers, and politeness (`[PLEASE]`, `[PLAIN]`). Every prompt includes the same legend, so all channels receive the same instructions. We count one action per chip, one per keyword under word prediction (or its characters without prediction), and one per suggestion

picked from a list. This is a deliberately simple proxy. Reading and checking suggestions costs real time [2, 36], and we do not measure effort, fatigue, or speed.

4.3. Prompts and models

All prompts describe the task as the sentence-composition feature of an AAC app. **P0** asks for "the single sentence they are trying to say" and decodes greedily. **S3** uses P0 with three samples at temperature 0.7, a natural multi-suggestion baseline. **P2** asks for "3 different suggestions that cover the different things the user might mean" as a JSON list and decodes greedily. For S3 and P2, each suggestion is scored on its own. We piloted a preservation prompt; it changed critical match by −3 to +7 points across channels on unvalidated pilot labels, at our pre-set kill line, so we dropped it before the main run to fund validation. We test three open models run locally with 4-bit quantization (Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct, Gemma-3-4B-it; Q4_K_M GGUF via llama.cpp) and one API model (gpt-6.1-terra). Local models reflect on-device deployment; we do not measure latency or privacy. Prompts, seeds, model hashes and every raw output are logged.

4.4. Label extraction: a judge that cannot see the answer

Judges never see the source message, the model, the prompt, or the channel. They see one output sentence and answer extraction questions with fixed options. The critical question asks what value the output expresses for the family's dimension, with "unclear/mixed" always available. The general questions cover polarity, speech act, agent, certainty and politeness. A second pass shows the visible input and asks whether the output adds a fact, reason, person, place or time that the input does not contain. Loss, Violation and Assumption are then computed deterministically from the extracted values, the item specification and the channel. Identical output texts are judged once, whatever model or condition produced them.

The judge is gpt-6.1-sol, a different model from the GPT model under test. Two prompt variants differ in wording, option order and item order; if they disagree, a cross-family arbiter (a Claude model) decides. Because judges can favor outputs from their own model family [34], we report the GPT generation arm's validation separately. The codebook fixes edge cases before scoring. "Can you…?" is a request without politeness, "Can I…?" is a question, and "I want X" has the speaker as agent. For certainty, an emphatic output matches a plain assertion and is logged as an added stance.

4.5. Human validation and adviser review

Two human coders labeled `[gold_n]` held-out outputs, using the same options and the same output-only view as the judge. The outputs were drawn with known inclusion probabilities, stratified by dimension, model and channel. Coders also labeled `[challenge_n]` authored challenge sentences: semantic mutations, such as a dropped negation or a swapped agent, and meaning-preserving paraphrases. We report Krippendorff's α between coders before adjudication, and Cohen's κ, sensitivity and specificity of the final judge pipeline against adjudicated labels. We use a dimension's judge labels only if its $\kappa \geq 0.6$; otherwise we report that dimension from human labels alone. Autistic advisers reviewed the codebook before the main run, focusing on what counts as added certainty, politeness or directness, and on the agent rules. [PENDING: date, number of advisers, changes accepted and rejected]

4.6. Statistics

Outputs within a family are not independent: two variants share one bare request, and three suggestions share one set. We therefore resample whole families (`[n_families_scored]` scored families), keeping all variants, channels, prompts and models of each sampled family together, and recompute every estimate and paired contrast in each of 2,000 replicates. We report 95% percentile intervals with equal family weights [37]. Corpus rates in S1 carry Wilson intervals over messages.

Table 3
S1 on AACText ([s1_n_messages] messages). Percentage of messages; 95% Wilson intervals in brackets. Detector-based estimates of cue-token loss, not semantic judgments.

Measure (POS-cap4 compressor)	% o
Loses ≥1 detected cue, any dimension	[s1_loss_any] [[s1_loss_any_lo], [s1_loss
excluding agent (pronoun) loss	[s1_loss_excl_agent] [[s1_loss_excl_agent_lo], [s1_loss_excl_a
excluding agent and time	[s1_loss_excl_ag
Loses a negation cue	[s1_loss_
Loses a speech-act cue	[s1_loss_sp
Keywords collide with a differently labeled message	[s1_c
All detected values restorable with ≤2 chips	[s1_res
Loss not expressible in the chip vocabulary	[s1_inexp

5. Results

5.1. Loss: keywords drop commitments in most messages

Content-word keywords lose at least one explicit cue in [s1_loss_excl_agent]% of AACText messages, even when we ignore pronouns (Table 3). Speech act is the most frequently lost dimension after agent and time: question and request forms are carried by function words and word order that the compressor removes. Negation is rarer but is always lost when present, because the compressor removes "not". In [s1_collision]% of messages, the keywords are identical to those of another message with a different commitment, so even a perfect expander could not recover both. Two chips or fewer would restore all detected values in [s1_restore_two]% of messages. This counts category values, not phrasing or negation scope. Loss is common, and cheap to mark in principle; what models do with the gap is a separate question.

5.2. Assumption: the unmarked reading is free

Under the bare channel, [bare_affirm_rate]% [[bare_affirm_rate_lo], [bare_affirm_rate_hi]] of definite polarity outputs were affirmative ([bare_affirm_n]/[bare_affirm_den]), across all four models and all prompts. The spanning prompt P2, which explicitly asks the model to cover different meanings, offered a negated suggestion in [bare_neg_offered_p2]% of polarity sets. Defaults appear on other dimensions as well, though they are less extreme. Bare outputs named the speaker as agent in [bare_speaker_rate]% of agent-family outputs, and used a plain assertion in [bare_plain_cert_rate]% of certainty-family outputs. They added "please" or a softener in [bare_please_added]% of outputs where the input had neither.

Adding the noncritical word changed the aggregate critical-match rate by [noncrit_vs_bare] points [[noncrit_vs_bare_lo], [noncrit_vs_bare_hi]]. It did change individual outputs ([noncrit_rows_differ]% of aligned rows), so we claim only that a longer input does not by itself move the aggregate default.

Figure 2 turns these rates into a cost. For a person who means the affirmative, the bare keywords already produce the intended polarity, at zero extra actions. For a person who means the negation, the bare keywords produce the opposite, and the intended meaning becomes available only after an extra selection. At the bare channel, the paired fidelity gap between the two directions was [action_gap_polarity] points [[action_gap_polarity_lo], [action_gap_polarity_hi]]. The minimum tested cost to reach 80% fidelity was [min_actions_aff] extra actions for affirmative variants and [min_actions_neg] for negated ones. The asymmetry is not a measurement artifact of our choice of base rate: the same bare input serves both variants, so any model that always picks one reading charges the other reading's speaker for every correction. Who pays is decided by the default, not by the person.

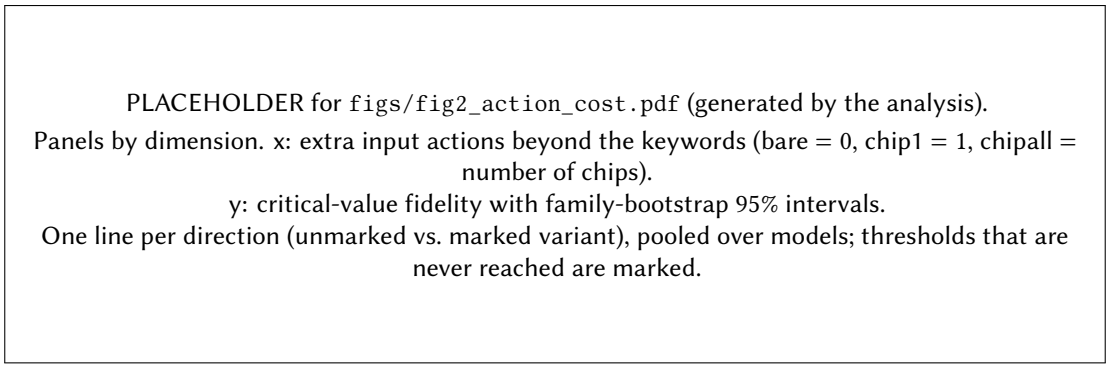


Figure 2: Fidelity against input actions, separately for each direction of each minimal pair. Under the bare channel the unmarked reading starts high and the marked reading starts low. One chip closes most of the gap. Intervals come from resampling families.

Table 4
Critical-value match (%) by dimension and channel, prompt P0, pooled over four models. For bare and noncrit the input is ambiguous, so match is at most the share of the more frequent variant and is not an accuracy.

Dimension	bare	noncrit	chip1	chipall	
Polarity	[match_pol_bare]	[match_pol_noncrit]	[match_pol_chip1]	[match_pol_chipall]	[match_pol_fu
Agent	[match_agt_bare]	[match_agt_noncrit]	[match_agt_chip1]	[match_agt_chipall]	[match_agt_fu
Certainty	[match_cer_bare]	[match_cer_noncrit]	[match_cer_chip1]	[match_cer_chipall]	[match_cer_fu
Speech act	[match_act_bare]	[match_act_noncrit]	[match_act_chip1]	[match_act_chipall]	[match_act_fu
Politeness	[match_pz_bare]	[match_pz_noncrit]	[match_pz_chip1]	[match_pz_chipall]	[match_pz_fu

5.3. Violation: cues mostly survive, explicit politeness choices do not

With one chip, polarity was preserved in [chip1_match]% [[chip1_match_lo], [chip1_match_hi]] of outputs. Across scored dimensions, one chip reached [chip1_match_all]% critical match, against [full_match_all]% when the model saw the full source (difference [chip1_vs_full] points [[chip1_vs_full_lo], [chip1_vs_full_hi]]; Table 4). In other words, a single selection recovers most of what the typed sentence provides on the tested dimension.

The full channel is the most instructive, because there the model sees exactly what the person typed. Models still changed it. For plain directives presented in full, [full_politeness_added]% [[full_politeness_added_lo], [full_politeness_added_hi]] of outputs added "please" or a softener ([full_politeness_added_n]/[full_politeness_added_den]). Fig. 1c shows one: "Wear headphones for your music." became "Please wear headphones for your music." Overall, [full_stance_changed]% of full-channel outputs changed at least one explicit value on a general dimension. Violations concentrated in speech act and politeness, not in polarity or agent. These changes look minor, and that is the point: an assistant tuned to be courteous treats a deliberately plain request as a draft to improve [28, 27].

5.4. Cue interaction: one more chip can undo a choice

Giving a model more commitments does not simply add up. In the pilot, [PLAIN] alone kept requests plain, but [ASK] [YOU] [PLAIN] led one model to add "please". On the first held-out politeness family, Gemma-3-4B turned [ASK] [YOU] [PLAIN] wear headphones into "Could you please tell me if you are wearing headphones?", which adds a softener and also changes the request into a question about the listener's state. We tested this effect directly (Experiment B). Holding [PLAIN] or [PLEASE] fixed on held-out politeness families, we crossed the presence of [ASK] and [YOU], and added a control with inert bracketed tags of matched length. Adding [ASK] and [YOU] lowered plainness preservation by [interaction_drop] points [[interaction_drop_lo], [interaction_drop_hi]],

against [interaction_padding_drop] points for the padding control, in [interaction_families] families. [PENDING: state which models show the effect; the pilot effect in Qwen disappeared once labels followed the codebook (see §5.6); report a model-specific counterexample if only one model shows it] Re-encoding the same commitments as transparent aliases or as short natural-language fields changed critical match by [encoding_alias_delta] and [encoding_nl_delta] points, respectively. In a factorial test that crossed polarity and actor cues on eight new families (Experiment C), adding the second cue lowered accuracy on the first by [factorial_protected_drop] points [[factorial_protected_drop_lo], [factorial_protected_drop_hi]]. Lower joint accuracy than either single cue is expected arithmetically; we report the drop on the already specified dimension, which is the quantity that matters to the person.

5.5. Suggestion sets: they span politeness, not polarity

Asking for three suggestions that cover different meanings (P2) covered both readings of a family in [coverage_p2]% [[coverage_p2_lo], [coverage_p2_hi]] of bare sets, against [coverage_s3]% [[coverage_s3_lo], [coverage_s3_hi]] for three sampled outputs (S3). The gain came mostly from politeness ([coverage_p2_pz]% vs. [coverage_s3_pz]%). For polarity, neither P2 nor S3 covered the negated reading ([coverage_p2_pol]% vs. [coverage_s3_pol]%). P2 suggestions added unlicensed content more often than S3 samples ([p2_additions]% vs. [s3_additions]%). Asking a model to diversify does not make it offer the reading that it never chooses by default.

5.6. Is the oracle trustworthy?

The two human coders agreed at α = [human_alpha] [[human_alpha_lo], [human_alpha_hi]] before adjudication. The final judge pipeline agreed with adjudicated labels at κ = [judge_kappa] [[judge_kappa_lo], [judge_kappa_hi]] on the critical dimension; the lowest per-dimension κ was [judge_kappa_min]. For Violation events, the judge's sensitivity was [judge_sens_violation]% and its specificity [judge_spec_violation]%; it labeled [challenge_acc]% of challenge sentences correctly. Validation changed conclusions, not just error bars. Our pilot's deterministic labeler counted "Can you…?" as a softened request, which made one model appear to lose plainness whenever [ASK] was added. Under the codebook rule, the same outputs are plain, and that apparent effect disappeared. Across headline contrasts, [stale_label_flips] changed by ten points or more between unvalidated and validated labels. [PENDING: if the validation gate fails for a dimension, say here which results rest on human labels only]

6. Discussion

Defaults are a policy choice. An expander that always picks the affirmative, the plain assertion, or the polite request has a policy, whether or not anyone wrote it down. The policy decides whose meaning arrives at zero cost and whose meaning needs an extra action. For a person who often refuses, negates, or speaks plainly, the system charges that person each time. How often any individual means the non-default reading, only they can say; but the designer sets the charge, and should do so on purpose and in the open.

Make non-default meanings cheap and visible. Commitment chips worked as an input channel: one selection recovered most of the full-source fidelity on the tested dimension. Chips need not be new widgets; intent buttons and stance tags already exist [13, 14, 15]. Our results suggest three refinements. First, place [NOT], [?] and [MAYBE] where they cost one selection, since these are the cues that the defaults override. Second, test combinations of chips, not only single chips, because one chip can undo another. Third, show the person which commitments the output makes, for example by highlighting "please" or a dropped hedge, so that a change is visible before the sentence is spoken.

Suggestion sets should span readings, not wordings. Where the input is ambiguous on a dimension that matters, such as polarity, a set should contain each reading, including the one the model would not choose on its own. Templates or explicit reading enumeration can guarantee this where prompting does not [30].

Never soften a plain directive the person typed. When the input already contains a complete sentence, the expander has nothing to expand. Adding "please" to "Wear headphones for your music" is a small edit, but it is the system speaking in the person's place. Some people may want an optional politeness assistant; that should be a setting they turn on, not a behavior they must notice and undo.

Limitations. We ran no study with AAC users, so we cannot speak to agency, effort, fatigue, or whether anyone wants a given output. AACText is fictional, crowd-written text, and our probes are constructed; neither shows what AAC users actually mean. Our cost model counts selections only and leaves out reading and checking suggestions, which can be substantial [36]. Commitments are text-explicit categories: a chip restores `[NOT]` but not its scope, and we did not score time and quantity. The judge is an LLM; we validated it against human labels on a limited sample, and dimensions below our agreement threshold rest on human labels alone. We tested four models, three of them small quantized models, with fixed prompts and no conversation context; larger models, context or personalization may shift the defaults. Ten families per dimension support estimates for these models and items, not population rates.

Autistic-informed, not essentialist. Autistic advisers shaped what we count as an added commitment, especially for directness, hedging and politeness. [PENDING: summarize their substantive changes] That review improves our definitions; it does not make our probes autistic speech, and it does not validate any claim that autistic people write in a particular way. We treat directness as one legitimate choice among others, to be kept when the person makes it, not as a trait to detect or correct.

7. Ethics and Positionality

This study collected no data from AAC users and no personal preference data. AACText is public under CC BY 4.0 and contains fictional messages [35]. Human coders labeled model outputs under a fixed codebook. [PENDING: institutional determination for the coding and adviser work; compensation of coders] [PENDING: Author positionality: state, as each author chooses, relevant lived experience (for example being autistic or using AAC) and disciplinary background, and how it shaped the work.]

Contribution of autistic advisers. [PENDING: Number of advisers, how they were recruited, when they joined (before the main run), what materials they received and in what format, accommodations offered (written or asynchronous participation, time to process), which decisions they could change, which changes we accepted and rejected and why, compensation, and how they chose to be credited (co-author, acknowledgment, or anonymous).] Following recent reviews of advisory practice in autism research [38], we describe this as a codebook review, not as an advisory board or co-design.

Release. [PENDING: Repository URL with probe families, codebook, prompts, raw outputs, judge prompts and labels, and analysis code.]

Acknowledgments

[PENDING: Acknowledgments, including advisers who chose to be named.]

Declaration on Generative AI

[PENDING: Replace with an accurate declaration. List every GenAI tool and service used and what each did, categorized with the CEUR GenAI Usage Taxonomy (for example literature search, code generation, drafting, figure scripting). Then state: "After using these tools/services, the authors reviewed and edited the content as needed and take full responsibility for the publication's content." Describe the models under test and the LLM judge as research instruments in the Method section, not here.]

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